

What the steric term does to $\partial c_i / \partial t$

A walk-through of eq. (15) of PNP Equation Formulations.tex

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Setup

The spec writes the modified Nernst–Planck equation (docs/PNP Equation Formulations.tex, line 159) as

$$\frac{\partial c_i}{\partial t} = \nabla \cdot \left\{ D_i \left[\nabla c_i + c_i \nabla \left(\frac{z_i F}{RT} \phi + \mu^{\text{steric}}(c) \right) \right] \right\}, \quad (1)$$

with the steric chemical potential at line 152

$$\mu^{\text{steric}}(c) = k_B T \ln(1 - \sum_j a_j c_j). \quad (2)$$

Two simplifications first:

- Define the local packing fraction $\Phi \equiv \sum_j a_j c_j$.
- Switch to the dimensionless drift potential $\tilde{\mu}^{\text{steric}} \equiv \mu^{\text{steric}} / (k_B T) = \ln(1 - \Phi)$. The units in (2) are sloppy: what actually goes inside the gradient in (1) has to be dimensionless (it is added to $z_i F \phi / RT$), so it is $\tilde{\mu}^{\text{steric}}$ that enters, not μ^{steric} as written. The qualitative story is identical.

The question we will answer is: *as c_i rises (and so Φ rises), what does the steric term do to $\partial c_i / \partial t$?*

1 Step 1 – write (1) as a drift–diffusion flux

Writing (1) as $\partial c_i / \partial t = -\nabla \cdot J_i$ gives

$$\begin{aligned} J_i &= -D_i \left[\nabla c_i + c_i \nabla \left(\frac{z_i F}{RT} \phi + \tilde{\mu}^{\text{steric}} \right) \right] \\ &= -D_i c_i \nabla \left[\ln c_i + \frac{z_i F}{RT} \phi + \tilde{\mu}^{\text{steric}} \right]. \end{aligned} \quad (3)$$

The flux is $-D_i c_i$ times the gradient of a total drift potential. The steric piece of that potential is just $\tilde{\mu}^{\text{steric}} = \ln(1 - \Phi)$.

2 Step 2 – gradient of the steric potential

Chain rule:

$$\nabla \tilde{\mu}^{\text{steric}} = \nabla \ln(1 - \Phi) = \frac{1}{1 - \Phi} \nabla(1 - \Phi) = -\frac{\nabla \Phi}{1 - \Phi}. \quad (4)$$

The steric *contribution* to the flux of species i is

$$\Delta J_i^{\text{steric}} = -D_i c_i \nabla \tilde{\mu}^{\text{steric}} = +D_i c_i \frac{\nabla \Phi}{1 - \Phi}. \quad (5)$$

Read the sign carefully. With the spec's $+\ln(1 - \Phi)$, the steric flux of species i points *up* the packing gradient – toward whichever side already has more counterion piled up. As $\Phi \rightarrow 1$ the prefactor $1/(1 - \Phi)$ diverges, so the up-gradient pump becomes infinitely strong. This is already wrong before we put it back into $\partial c_i / \partial t$.

3 Step 3 – collapse to a single-species effective diffusivity

To see what (1) actually does to $\partial c_i / \partial t$ as c_i grows, switch off electromigration for a moment and assume one dominant counterion, so $\Phi = a_i c_i$ and $\nabla \Phi = a_i \nabla c_i$. Combining the Fickian term and (5),

$$\begin{aligned} J_i &= -D_i \nabla c_i + D_i c_i \frac{a_i \nabla c_i}{1 - a_i c_i} \\ &= -D_i \nabla c_i \left[1 - \frac{a_i c_i}{1 - a_i c_i} \right] \\ &= -D_i \nabla c_i \cdot \frac{1 - 2\Phi}{1 - \Phi}. \end{aligned} \quad (6)$$

So (1), with the spec sign and electromigration off, collapses to a nonlinear heat equation

$$\boxed{\frac{\partial c_i}{\partial t} = \nabla \cdot (D_{\text{eff}}(\Phi) \nabla c_i), \quad D_{\text{eff}}(\Phi) = D_i \frac{1 - 2\Phi}{1 - \Phi}.} \quad (7)$$

That $D_{\text{eff}}(\Phi)$ is the whole story.

4 Step 4 – what D_{eff} does as c_i grows

Tabulate (7) versus packing fraction:

$\Phi = a_i c_i$	$D_{\text{eff}}(\Phi)$	what eq. (15) does
0	D_i	ordinary Fickian diffusion
$0 < \Phi < 1/2$	$0 < D_{\text{eff}} < D_i$	diffusion damped – already a small artifact
$\Phi = 1/2$	0	diffusion freezes – no spatial relaxation
$1/2 < \Phi < 1$	negative	backward heat equation – ill-posed
$\Phi \rightarrow 1$	$-\infty$	runaway: gradients amplified arbitrarily fast

The pathology starts at $\Phi = 1/2$, *not* at $\Phi = 1$. Anywhere the counterion piles past half-packing, the spec's eq. (15) is locally a backward heat equation: the diffusive term *amplifies* small bumps in c_i instead of smoothing them. There is no steady state for it to find – and that is exactly why Newton cannot solve the 4sp warm-walk above $\psi_D \approx 4.83$ (nondim, $\approx +0.124$ V vs RHE), which is where the ClO_4^- EDL crosses $\Phi \approx 0.5$. The solver is not failing for numerical reasons – the PDE itself stops being well-posed.

5 Step 5 – contrast with the corrected sign $-\ln(1 - \Phi)$

Repeat Steps 2–3 with $\tilde{\mu}^{\text{steric}} = -\ln(1 - \Phi)$ (Borukhov–Andelman–Orland 1997 eq. (3); Bazant–Kilic–Storey–Ajdari 2009 eq. (20)). Then $\nabla \tilde{\mu}^{\text{steric}} = +\nabla \Phi / (1 - \Phi)$ and

$$\begin{aligned} J_i &= -D_i \nabla c_i - D_i c_i \frac{a_i \nabla c_i}{1 - a_i c_i} \\ &= -D_i \nabla c_i \left[1 + \frac{a_i c_i}{1 - a_i c_i} \right] \\ &= -D_i \nabla c_i \cdot \frac{1}{1 - \Phi}. \end{aligned} \tag{8}$$

Equivalently,

$$D_{\text{eff}}(\Phi) = \frac{D_i}{1 - \Phi} \quad (\text{Borukhov sign}). \tag{9}$$

Always positive, monotonically growing, and $\rightarrow +\infty$ as $\Phi \rightarrow 1$. Diffusion gets *stronger* the closer the layer is to saturation, which is precisely what enforces the Bikerman cap $c_i \leq 1/a_i$ (Borukhov 1997 eq. (5)).

Summary

- With the spec sign $+\ln(1 - \Phi)$, the steric piece of the Nernst–Planck flux pumps mass *up* the packing gradient. In the single-species reduction this gives $D_{\text{eff}}(\Phi) = D_i(1 - 2\Phi)/(1 - \Phi)$, which is negative for $\Phi > 1/2$. Eq. (15) then becomes a backward heat equation in the saturation skin – ill-posed, no steady state, Newton diverges.
- With the corrected Borukhov/Bazant sign $-\ln(1 - \Phi)$, the steric piece of the flux pushes mass *down* the packing gradient. The single-species reduction gives $D_{\text{eff}}(\Phi) = D_i/(1 - \Phi)$, monotonically growing and diverging at saturation. The PDE is well-posed and produces the Bikerman cap $c_i \leq 1/a_i$.

Caveats. Two simplifications I papered over: (i) the cross-species term in $\Phi = \sum_j a_j c_j$ produces off-diagonal $c_i \nabla c_j / (1 - \Phi)$ contributions that do not show up in the single-species reduction, but they have the same sign structure and do not rescue the spec form; (ii) electromigration adds $c_i z_i F \nabla \phi / RT$ to the flux, which does not enter D_{eff} . It only shifts *where* Φ first crosses $1/2$ – it does not change that the equation goes ill-posed once it does.

Where the fix lives in code.

- Forward/bv_solver/forms_logc.py:303 — `mu_steric = -fd.ln(packing)`
- Forward/bv_solver/forms_logc_muh.py:347 — same.
- Gated by `steric_active` (line 283 / 335), which is `True` only when some $a_i \neq 0$ or a Bikerman counterion is configured. The 3sp+Boltzmann ideal preset ($a_{\text{vals_hat}} = [0, 0, 0]$, no `steric_mode='bikerman'`) is unaffected.

The code is now consistent with the corrected sign; the spec doc at `docs/PNP Equation Formulations.tex:152` still has the original sign and disagrees with the code. The plan in `docs/steric_sign_correction_plan.md` flags this as needing explicit advisor sign-off before the spec edit ships.